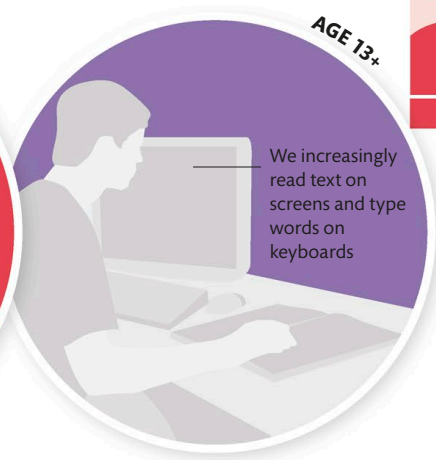
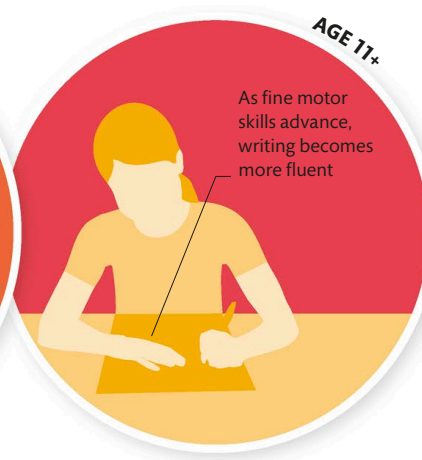




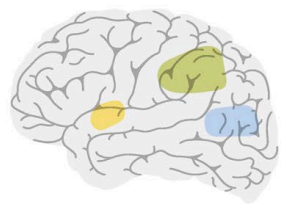
3 Beginning to read
Reading aloud can improve a child's reading ability. Listening to a story activates the auditory cortex to hear the words, which are then processed by the frontal lobe. Picture books help children practice relating words to images, and asking them to join in reading builds vocabulary and comprehension.

4 Expanding vocabulary
As we grow older, we experience more of the world around us so we learn and see new things, adding to our vocabulary. Comprehension, the ability to understand how to use words, requires every lobe of the brain (see p.30) and the cerebellum to successfully comprehend and use language.

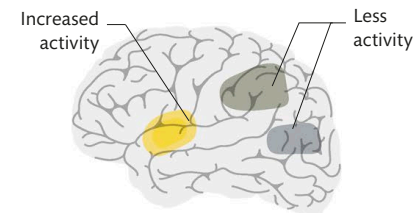
5 Continuing to learn
As adults, we continue to learn and practice our reading and writing skills. Our vocabulary is constantly being extended. Learning to read and write is just the start of the story. The whole brain is required to maintain language skills, and good brain health is vital to both reading and writing.

Dyslexia

Dyslexia takes various forms, affecting people's ability to read or write, or both. It is thought that up to one in five people have dyslexia. A full neurological explanation of the causes of dyslexia has not yet been achieved. Studies have suggested that particular structures of the brain function differently in dyslexia (see right). As children with dyslexia typically struggle with their reading abilities, it is difficult to determine whether the developing brain impacts the dyslexia or if the dyslexia itself has an impact on the developing brain.



Nondyslexic brain reading
Broca's area helps form and articulate speech. The parietal-temporal cortex works to analyze and understand new words. The occipital-temporal area forms words and aids in meaning, spelling, and pronunciation.



Dyslexic brain reading
Broca's area is activated to form and articulate words, but the parietal-temporal and occipital-temporal areas are less active. Broca's area can be overactivated to compensate for the lack of stimulation of the other regions.

KEY

- Parietal-temporal
- Inferior frontal gyrus (Broca's area)
- Occipital-temporal

ALPHABETIC PRINCIPLE

The alphabetic principle is the idea that individual letters or groups of letters represent sounds when they are spoken aloud. The alphabetic principle has two parts:

- 1. Alphabetic understanding**
Learning that words are made up of letters that represent the sounds made when speaking these letters aloud.
- 2. Phonological recoding**
Understanding how strings of letters in written words combine to make sounds, which enables spelling and pronunciation.

